

Stuti Shah

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PROFESSIONAL EXPERIENCE

Software Engineering Intern, Volvo Group Greensboro, NC	May 2025 - Aug 2025
- Built distributed backend systems using Python, Flask, PostgreSQL, and Kubernetes, improving reliability and scalability of engineering services. - Designed automated engineering pipelines that reduced workflow execution from 4 hours to 6 minutes, improved infrastructure efficiency, and generated \$176,980 in annual operational savings. - Developed high-throughput REST services processing large-scale vehicle test data with sub-second query performance. - Deployed services through GitOps-based infrastructure (Argo CD + Kubernetes), ensuring reproducible environments.	
AI Intern, AquaOrange Bangkok, Thailand	Jun 2024 - Aug 2024
- Automated AI-driven test case generation for CRM systems, enhancing defect detection by 20%. - Utilized AI testing tools for frontend and backend validation, reducing regression risk and improving stability. - Integrated automated testing into CI pipelines, accelerating issue detection and enhancing release reliability.	
Solutions Associate, HERE! Wireless USA Dallas, TX	Jun 2023 - Jul 2024
- Conducted comprehensive alpha testing and root-cause analysis for Android and iOS apps, improving parental control feature stability and reducing critical crash reports by 35%. - Collaborated with product managers and engineers to translate customer feedback into actionable engineering requirements, accelerating feature refinement and enhancing user satisfaction across 200+ active users. - Performed structured analysis of pricing, usage, and customer behavior data, generating insights that informed product decisions and optimized feature adoption, reducing churn risk by measurable margins.	
CS Undergraduate Teaching Assistant, Virginia Tech Blacksburg, VA	Aug 2024 - May 2025
- Mentored 200+ students in computer architecture and low-level systems, covering processor pipelines, assembly execution, and memory hierarchy. - Assisted students with assembly programming and performance analysis, debugging memory access patterns and instruction-level execution. - Guided coursework involving processor architecture, caching behavior, and system-level performance optimization.	

EDUCATION

Virginia Tech - Master of Engineering, Computer Science	May 2026
Virginia Tech - Bachelor's, Computer Science	May 2025

PROJECTS

CS Source JobHub
Developed secure backend services and REST APIs using Flask and MySQL, supporting role-based access, membership lifecycle management, and scalable workflows with strong data integrity guarantees.
YOLOv8 Shelf Detection for Planogram Compliance (Items + Empty Spaces)
Built an end-to-end shelf monitoring pipeline using YOLOv8 to detect products and empty shelf spaces from aisle images, curating/training on public datasets to support planogram compliance and restocking insights.
HybridDiff AR: Hybrid Diffusion-Autoregressive Model for Efficient and High-Quality Image Generation
Developed a hybrid image generation pipeline, merging a 33M parameter DDPM with transformer refinement for 250× faster inference on CelebA 64×64.
Location-Aware Task Management App
Built an MVVM-based Android application using Room, LiveData, and ViewModel, enabling offline-first CRUD workflows and location-aware task execution with predictable, low latency user interactions.

SKILLS

Programming Languages : C, Python, Java
Systems Programming & Operating Systems : Linux, Process & Thread Management, Concurrency & Synchronization, Interprocess Communication (IPC), System Calls, Memory Management, Virtual Memory
Computer Architecture : Processor Architecture, Instruction Pipelines, Memory Hierarchy (Cache, Paging), Assembly-Level Execution, Performance Optimization
Distributed Systems & Backend : REST APIs, Distributed Systems, Microservices, Authentication & Authorization, RBAC
Databases & Storage Systems : PostgreSQL, MySQL, Relational Schema Design, Query Optimization, Transaction Management
Cloud & DevOps : Docker, Kubernetes, GitOps (Argo CD), CI/CD Pipelines, Linux Environments
Software Engineering : Data Structures & Algorithms, Object-Oriented Programming, Unit Testing, Logging & Monitoring